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BSIT 301I

Conceptual Framework

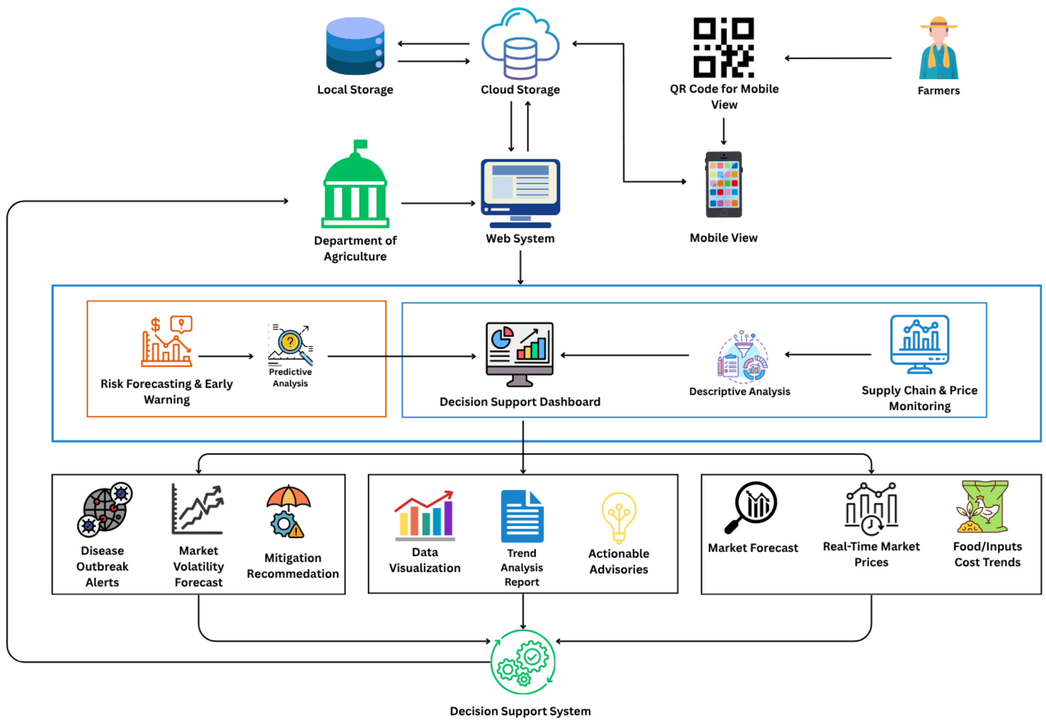
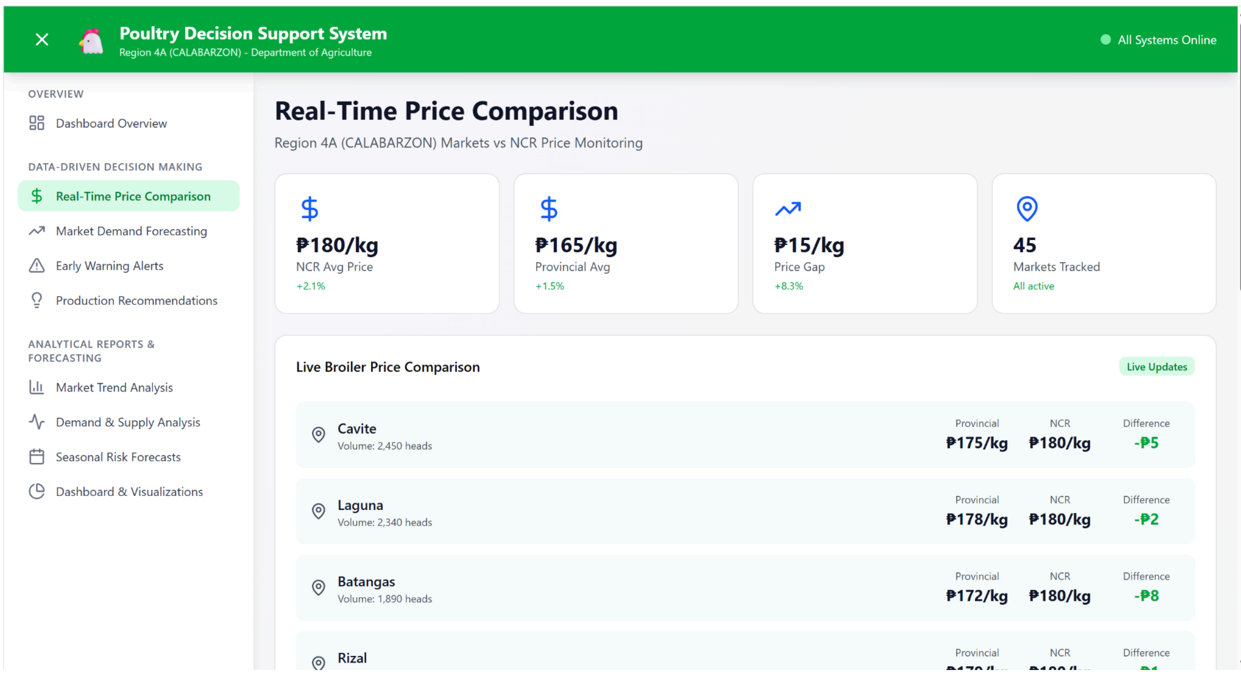


Figure 1.1 Conceptual Framework of Proposed Study

The conceptual framework illustrates the cyclical data flow and architecture of the proposed Decision Support System and Risk Forecasting tool designed for poultry and egg stakeholders and the Department of Agriculture (DA) in Region IV-A. The process begins when farmers access the system by scanning a QR code, allowing them to input real-time farm data such as production levels, feed consumption, environmental status, and mortality rates. This information is instantly transmitted through a secure cloud database—which is also synced to local storage for backup—directly to the DA's web system. Acting as the central administrator, the DA combines the farmers' information with macro-level inputs, including regional policies, market and supply chain statistics, reported disease outbreaks, and environmental changes. The DA then processes this integrated data using three core subsystems: Decision Support, Risk Forecasting & Early Warning, and Supply Chain Monitoring. To analyze the data, the system relies on descriptive analytics using a Simple Moving Average (SMA) approach to smooth out short-term fluctuations and identify clear historical trends in egg prices, feed costs, and production levels. Furthermore, the system employs Time-Series analysis for predictive analytics; by examining time-based data and seasonal variations, the Time-Series models can accurately forecast future market dynamics, impending disease outbreaks, and the potential impacts of environmental shifts. Once the information is processed, the system completes the communication loop by sending targeted, actionable insights back to the farmers' mobile view. Farmers receive real-time price updates, revenue profitability metrics, and a visual dashboard tracking feed efficiency and poultry loss, along with critical disease and climate alerts. Simultaneously, the DA receives aggregated regional market analytics and disease outbreak forecasts, empowering them to proactively generate and disseminate data-driven advisories to help farmers mitigate ongoing agricultural challenges.

3 Subsystem Prototype



Poultry Decision Support System
Region 4A (CALABARZON) - Department of Agriculture
All Systems Online

	Provincial	NCR	Savings
Laguna Medium Size	₱225	₱235	-₱10
Batangas Large Size	₱240	₱255	-₱15
Quezon Medium Size	₱210	₱235	-₱25

Price Arbitrage Opportunities

Quezon to NCR — High Volume
Product: Live Broiler
Potential Margin: ₱12/kg

Cavite to NCR — Medium Volume
Product: Eggs (Medium)
Potential Margin: ₱15/tray

Batangas to NCR — Medium Volume
Product: Live Broiler
Potential Margin: ₱8/kg

7-Day Price Gap Trend

Today	₱15/kg	+2.1%
Yesterday	₱14.5/kg	+1.8%
2 days ago	₱14/kg	+0.5%
3 days ago	₱13.8/kg	-0.3%
4 days ago	₱14/kg	+1.2%
5 days ago	₱13.5/kg	+0.8%
6 days ago	₱13.2/kg	0%

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- Dashboard Overview

DATA-DRIVEN DECISION MAKING
- Real-Time Price Comparison
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Market Demand Forecasting

Historical data analysis and demand predictions for Region 4A (CALABARZON)

+12%	91.5%	24K	15 days
30-Day Forecast	Forecast Accuracy	Data Points Used	Next Peak
Increasing	High	36 months	Holiday Season

30-Day Demand Forecast by Product

Live Broiler — High (92%)
+13.3% Projected Growth
Current Demand: 45,000 heads/day
30-Day Forecast: 51,000 heads/day

Key Drivers:
- Holiday season approaching
- Restaurant demand increasing

Eggs (Medium) — High (89%)
+12.5% Projected Growth

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Historical Demand Patterns

Seasonal Trends

December Peak:	+35% demand
June-July Dip:	-12% demand
Weekend Spike:	+18% avg

Monthly Comparison

January	Actual: 42,500	98%
February	Actual: 41,800	99%
March	Actual: 44,200	99%

Demand by Market Segment

Retail/Wet Markets 45%
Current Volume: 22,500 heads/day
Forecast Growth: +10%

Supermarkets 25%
Current Volume: 12,500 heads/day
Forecast Growth: +15%

Food Service/Restaurants 20%
Current Volume: 10,000 heads/day
Forecast Growth: +18%

Industrial/Processing 10%
Current Volume: 5,000 heads/day
Forecast Growth: +8%

Production Recommendations Based on Forecast

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Early Warning Alert System

Disease outbreaks and seasonal risk monitoring for Region 4A (CALABARZON)

2	5	38	12/week
Critical Alerts	Active Warnings	Monitored Areas	Alert Rate
High	Medium	All regions	Avg

Active Disease Outbreak Alerts

2 Critical

Avian Influenza (H5N1) Critical
Batangas - Lipa City
Detected: 2 days ago

Affected	Status	Spread Risk	Radius
3 farms, ~15,000 birds	Quarantine Active	High risk	10 km

Response Actions:
Immediate culling | 10km radius quarantine | Movement restrictions

Newcastle Disease Critical
Laguna - Calamba
Detected: 4 days ago

Affected	Status	Spread Risk	Radius

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Seasonal Risk Warnings

Heat Stress Risk — High Risk
Summer Peak (April-May)
Impact:
Reduced feed intake, lower egg production, mortality risk
Prevention:
Improve ventilation, provide cooling, adjust feeding schedule

Rainy Season Diseases — Medium Risk
Monsoon (June-October)
Impact:
Increased respiratory diseases, higher humidity diseases
Prevention:
Enhanced biosecurity, drainage management, dry litter maintenance

Cold Stress in Chicks — Medium Risk
Cool months (December-February)
Impact:
Chick mortality, growth retardation, immune suppression
Prevention:
Brooder temperature control, draft prevention, supplemental heating

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Risk Level by Region

Batangas — Critical
Active diseases: Avian Influenza
Affected farms: 3

Laguna — High
Active diseases: Newcastle Disease
Affected farms: 2

Cavite — Medium
Active diseases: Infectious Bronchitis
Affected farms: 1

Rizal — Low
Active diseases: None active
Affected farms: 0

Quezon — Low
Active diseases: None active
Affected farms: 0

Alert History (Last 30 Days)

Alert Frequency

Week 1	**2 alerts**	1 Critical, 1 Medium
Week 2	**4 alerts**	1 Critical, 3 Low
Week 3	**3 alerts**	2 Medium, 1 Low
Week 4	**5 alerts**	2 High, 3 Medium

Trending Risk
Avian Influenza cases detected in CALABARZON region
Recommendation: Enhance biosecurity measures in all farms within 50km radius

Success Rate
87%
Early detection and containment success rate

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Production Recommendations

Data-driven insights for Region 4A (CALABARZON) based on market trends and risk analysis

12
Active Recommendations

4
High Priority

28
Implemented

+18%
Potential ROI

High Priority Recommendations

Increase Broiler Production by 15% High Priority

Based on: Market Trend Analysis + Demand Forecasting

Analysis & Reasoning:
Holiday season demand surge expected in 15 days. Current production insufficient to meet projected 51,000 heads/day demand.

Timeline	Potential Impact	Confidence	Risk Level
Start immediately - 2 week ramp-up	+₱2.4M revenue potential	92%	Feed cost increase

Action Steps:
- Order additional day-old chicks (DOCs) - 15,000 units
- Secure feed supply for 6 weeks
- Prepare grow-out facilities

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Medium Priority Recommendations

Diversify into Organic/Free-Range Production Medium Priority

Market trend showing 25% growth in premium segment

Potential	Timeline	Confidence
+₱350K potential	3-6 months	72%

Establish Direct Supply to Restaurants Medium Priority

Food service demand +18%, better margins than wet market

Potential	Timeline	Confidence
+₱280K margin improvement	1-2 months	68%

Implement Feed Cost Optimization Medium Priority

Feed costs up 8%, alternative formulations available

Potential	Timeline	Confidence
Save ₱450K annually	Immediate	85%

Expand Egg Production Capacity Medium Priority

Sustained egg demand growth, stable prices

Potential	Timeline	Confidence
+₱520K revenue	6-8 months	75%

Risk Mitigation Recommendations

Implementation Success Tracking

Risk Mitigation Recommendations

Critical: Avian Influenza Prevention

AI outbreak in Batangas poses regional risk across CALABARZON

- Implement strict biosecurity protocols
- Limit farm visits and vehicle movement
- Prepare emergency response plan
- Consider production insurance

Heat Stress Management

Summer peak approaching, production impact expected

- Install cooling systems now
- Adjust feeding schedules
- Ensure water availability

Feed Supply Chain Resilience

Feed costs volatile, supply disruptions possible

- Secure 60-day feed inventory
- Diversify feed suppliers
- Explore alternative ingredients

Implementation Success Tracking

Recently Implemented

Market diversification	+15% revenue	Success
Feed optimization	8% cost reduction	Success
Quality improvement	+₱12/tray premium	Success

Overall Impact

+22% Revenue Growth

89% Success Rate

Next Review

Recommendations updated weekly based on:

- Real-time market price data
- Demand forecast updates
- Disease outbreak monitoring
- Seasonal risk assessments

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Seasonal Risk Forecasts

Disease occurrence trends and seasonal risk patterns for Region 4A (CALABARZON)

Dry	**Medium**	**42/100**	**82%**
Current Season	Risk Level	Disease Index	Historical Pattern
Feb-May	Heat stress	Moderate	Accuracy

Annual Seasonal Risk Calendar

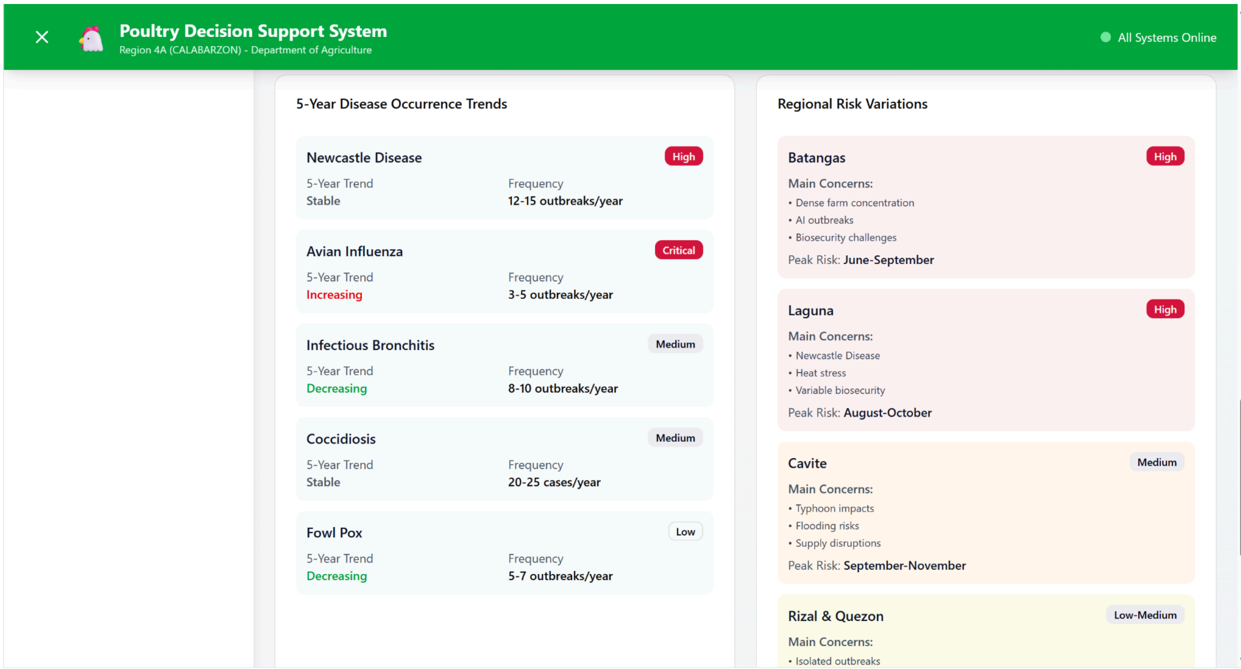
Dry Season (February - May) — Medium Risk

Feb, Mar, Apr, May

Primary Risks:

Heat Stress	Impact: Production decline 15-25%	High (85%)
Dehydration	Impact: Mortality increase 5-8%	Medium (45%)
Dust-related Respiratory Issues	Impact: Feed conversion down 10%	Medium (38%)

Preventive Measures:



The three subsystem prototypes embody the fundamental functionalities of the envisioned Poultry Decision Support System, each tailored to assist both poultry farmers and the Department of Agriculture in Region IV-A. These subsystems are integrated to furnish prompt market intelligence, forecast potential risks, and formulate well-informed recommendations, thereby facilitating improved decision-making processes.

The first subsystem, Supply Chain and Price Monitoring, is dedicated to monitoring current market dynamics and conducting price comparisons across various geographical areas.

This module allows stakeholders to view current price levels, market differences, price trends, and possible arbitrage opportunities. By consolidating supply chain and pricing data in a single interface, the subsystem helps farmers identify potentially more profitable sales opportunities and allows the DA to monitor market changes more effectively.

The second subsystem, Risk Forecasting and Early Warning, is designed to identify potential threats before they significantly harm poultry production. This includes monitoring disease outbreaks, seasonal risks, and other environmental factors that could disrupt farming operations. The module presents alert levels, regional risk status, and seasonal disease patterns, while also recommending preventive actions such as biosecurity measures and mitigation strategies. This feature provides stakeholders with advanced notice, potentially allowing them to mitigate losses and bolster their readiness.

The Decision Support Dashboard, the system's third subsystem, serves as its central analytical and advisory component. This element synthesizes data from market, supply chain, and risk forecasting modules to generate user-specific, actionable insights. Consequently, the dashboard offers production recommendations, trend analyses, and performance metrics, thereby assisting farmers in optimizing production levels, managing feed consumption, and improving operational planning. This subsystem effectively converts raw data into practical guidance, thereby facilitating both immediate and strategic decision-making processes.

Conclusion

The conceptual framework demonstrates a comprehensive and iterative information exchange between farmers and the Department of Agriculture. Farmers use the mobile interface to enter real-time data about their farms. The DA then combines this information with regional market data, disease reports, and environmental details. The data then gets processed by the system through its three main components: Supply Chain and Price Monitoring, Risk Forecasting and Early Warning, and the Decision Support Dashboard. The actionable insights and recommendations emerge from this process.

The framework shows that the proposed system is not only for recording data, but also for analyzing trends, forecasting risks, and supporting decisions in the poultry and egg industry. By using descriptive analytics, Simple Moving Average, and Time-Series analysis, the system can help stakeholders better understand market changes, anticipate possible problems, and respond more effectively.